Lesson 6: Working with Multiple Objects Using Arrays or Lists

Focus: Creating and managing collections of objects.

The school now wants the system to manage an entire class rather than individual students.

Exercise 6 – Managing a Class of Students

Continue using the `Student` class.

Create a list or array called `students` that will store multiple Student objects.

Questions

1. Create an empty list or array that can store Student objects.
2. Create five student objects with different information and examination marks.
3. Add all five objects to the collection.
4. Use a loop to display the details of every student.
5. Use a loop to calculate and display the average mark of each student.
6. Identify the student with the highest average mark.
7. Identify the student with the lowest average mark.
8. Count how many students passed and how many failed.
9. Calculate the overall class average.
10. Display a summary showing:
    - Total number of students.
    - Number of students who passed.
    - Number of students who failed.
    - Highest student average.
    - Lowest student average.
    - Overall class average.

Challenge question: Sort the students according to their average marks, starting with the highest-performing student.

Learning checkpoint: Learners should understand that objects can be stored in arrays or lists and processed using loops.